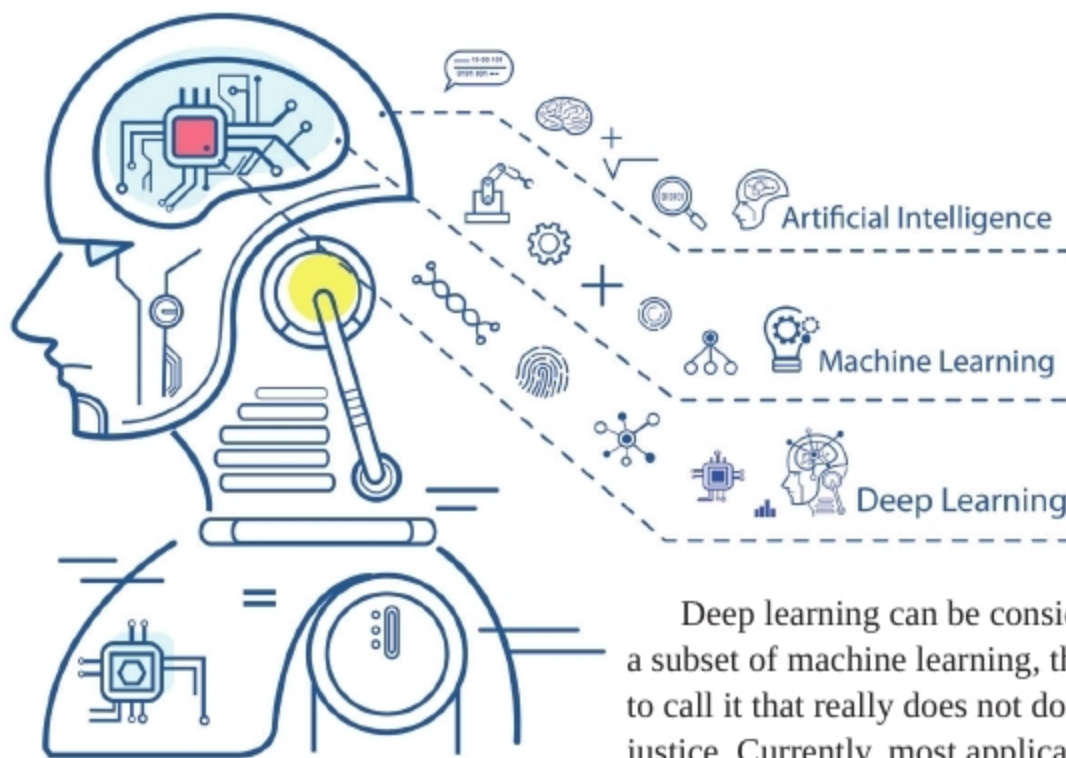


Tools that Accelerate a Newbie's Understanding of Machine Learning

The world of machine learning (ML) and deep learning (DL) is a fascinating one. Many newbies would like to dabble in this field but have some inhibitions. This article introduces readers to some of the best tools to kickstart their journey into the ML/DL domain.



Machine learning is a term that is applied to a broad range of topics which have one thing in common – the use of algorithms and other statistical models to improve the performance of a particular task. All machine learning (ML) models are used to build a mathematical model of the data provided, to help the user predict or make decisions with a very high level of accuracy; thereby considerably reducing the strain of manually sifting through the data. Currently, there are several fields in which ML is used extensively, from filtering out emails and computer vision, to choosing a smart assistant. Readers are urged to spend some time researching only the topics that they are interested in and not get overwhelmed or distracted by the countless other applications of ML.

Deep learning can be considered a subset of machine learning, though to call it that really does not do it justice. Currently, most applications have moved on from the somewhat old fashioned ML algorithms to employ deep learning (DL) algorithms. The latter provide much more support for the building of more complex mathematical models which cannot be provided by ML. Deep learning algorithms employ multiple ML algorithms in a manner vaguely inspired by how the human brain works, and hence the term 'neural networks'. Deep learning algorithms have been developed to the point where it has been proved that computers can indeed be smarter than humans.

This article explores five of the most user-friendly, highly scalable ML libraries/tools available. These are:

- Scikit-learn
- OpenCV
- TensorFlow
- Keras

▪ Google Colaboratory

Some of you may be familiar with a couple of these libraries and even know how to use them. This is not meant to be an in-depth tutorial for any of these tools. The idea is that, hopefully, by the end of this article, readers will discover which of the five libraries featured, interests them enough to read further on the topic.

Scikit-learn

The founders of Scikit-learn started off by trying to find answers to simple problems. I believe that any aspirant must follow the pioneer's footsteps. Scikit-learn is a free library written for Python which lets users do exactly that. It operates using Python's NumPy and SciPy libraries to achieve multiple aspects of machine learning with ease, such as various algorithms for classification, regression and clustering. It lets the user easily step into the world of artificial intelligence (AI) without making them feel that they've taken too big a leap. With its easy-to-use libraries, you can easily incorporate any of the above ML algorithms into your code. While Scikit-learn is essentially for older ML algorithms, beginners will have their hands full trying to implement various algorithms to see how they affect a particular use case.

This library is largely implemented in Python with some parts of it written in Cython, thus restricting its usage to